CAPA: Structure-Aware HLA Mismatch Representations with Protein Language Models for Competing-Risks Survival Prediction in Haematopoietic Stem Cell Transplantation

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Abstract

Motivation: Allogeneic haematopoietic stem cell transplantation (HSCT) is curative for many haematological malignancies, but post-transplant outcomes remain unpredictable. Current donor selection relies on categorical HLA match/mismatch counts that discard allele-level structural information and ignore the competing nature of graft-versus-host disease (GvHD), relapse, and transplant-related mortality (TRM).

Results: We present CAPA (Computational Architecture for Predicting Alloimmunity), a deep learning framework that replaces categorical mismatch scores with continuous, biologically meaningful representations of HLA allele pairs. CAPA encodes each allele as a 1280-dimensional vector via the frozen ESM-2 650M protein language model, then learns donor–recipient interaction features through bidirectional cross-attention. A DeepHit survival head jointly models all three competing events over a 730-day horizon. On the public UCI Bone Marrow Transplant dataset ($n = 187$ paediatric patients), CAPA achieves time-dependent concordance indices of 0.68 ± 0.04 (GvHD), 0.65 ± 0.05 (relapse), and 0.70 ± 0.04 (TRM), outperforming a cause-specific Cox model, a Fine–Gray subdistribution model, and a random survival forest across all three end-points. Cross-attention weights recover known immunological hierarchies, assigning highest interaction strength to the class II loci DRB1 and DQB1.

Availability and Implementation: Source code, pre-trained weights, and a web interface are freely available at `https://github.com/capa-project/capa` under the MIT licence.

Keywords: HLA; protein language model; competing risks; haematopoietic stem cell transplantation; graft-versus-host disease; deep learning; survival analysis; cross-attention

1 Introduction

Allogeneic haematopoietic stem cell transplantation (HSCT) offers the only curative option for many haematological malignancies including acute leukaemias, myelodysplastic syndromes, and aplastic anaemia [2]. The central immunological determinant of transplant success is the degree of compatibility between human leucocyte antigen (HLA) molecules of the donor and recipient. HLA molecules (class I: HLA-A, -B, -C; class II: HLA-DRB1, -DQB1) present peptide antigens to T cells; even single amino-acid differences between donor and recipient alleles can provoke life-threatening alloreactivity [23, 29].

Current clinical practice scores donor–recipient compatibility by counting the number of mismatched loci at high-resolution (allele-level) typing, with 8/8 or 10/10 matching at HLA-A, -B, -C and -DRB1 (and optionally -DQB1) considered optimal [17]. This categorical approach has two fundamental limitations. First, it treats all mismatches as immunologically equivalent regardless of the physicochemical distance between allele products—mismatching at two alleles that encode structurally similar peptide-binding grooves may be far less harmful than mismatching at two alleles that differ at immunodominant positions [5, 24]. Second, it assigns a single aggregate score to pairs that may differ dramatically in which specific loci are mismatched, losing information that may be critical for predicting the risk of individual outcomes.

Protein language models (PLMs) offer a principled solution to the first limitation [8, 18, 25]. Trained by masked-language modelling on hundreds of millions of evolutionary sequences, PLMs learn residue-level embeddings that implicitly encode three-dimensional structure, functional constraints, and evolutionary conservation. Applied to HLA allele protein sequences, PLM embeddings capture the continuously-variable immunological distance between alleles in a way that categorical match/mismatch flags cannot.

The second limitation—modelling the competing nature of post-transplant events—requires methods that go beyond standard time-to-event regression. GvHD, relapse, and TRM are *competing risks*: the occurrence of one event precludes or alters the probability of the others, so independent cause-specific models are statistically incorrect [9]. The DeepHit framework [16] addresses this by learning a joint discrete-time distribution over event types and times, producing calibrated cumulative incidence functions (CIFs) for each competing outcome.

Here we introduce CAPA, which integrates all three components: (i) ESM-2 1280-dimensional HLA allele embeddings, (ii) bidirectional cross-attention to model locus-level donor–recipient interactions, and (iii) a DeepHit survival head for joint competing-risks prediction. We evaluate CAPA on the publicly available UCI Bone Marrow Transplant (BMT) paediatric dataset [26], demonstrating consistent improvement over classical and machine-learning baselines and presenting an attention-based interpretability analysis that aligns with established immunobiology.

2 Methods

2.1 Dataset

We used the UCI Bone Marrow Transplant: Children dataset [26, 27], comprising 187 paediatric patients who underwent allogeneic HSCT between 1992 and 2004 at a single centre. Each record includes donor and recipient HLA typing at loci A, B, C, DRB1, and DQB1 (allele-level resolution for the primary analyses), seven clinical covariates (recipient age, donor age, disease category, conditioning regimen, donor type, graft source, sex mismatch), and three time-to-event outcomes: acute GvHD (grade II–IV), relapse, and transplant-related mortality (TRM). Survival time was censored at the last follow-up visit or day 730 post-transplant, whichever occurred first.

Data were split into training, validation, and test sets (70 / 15 / 15 stratified by event type) using a competing-risks-aware stratification procedure that preserves marginal event-rate distri-

butions across splits. All continuous covariates were standardised to zero mean and unit variance using training-set statistics. Missing HLA fields (2.1 % of locus–subject pairs) were imputed with zero-vectors at the embedding level.

2.2 HLA Sequence Representation with ESM-2

HLA allele protein sequences were retrieved from the IPD-IMGT/HLA database (release 3.55.0) [4]. Each allele was mapped to its full extracellular amino-acid sequence (typically 274 residues for class I, 194 for class II beta chains).

Sequence embeddings were computed using the pre-trained ESM-2 650M parameter model (`facebook/esm2_t33_650M_UR50D`) [18]. We extracted the per-residue hidden states from the final transformer layer and averaged them to obtain a single 1280-dimensional allele embedding. ESM-2 weights were kept fully frozen throughout all experiments; no fine-tuning was performed. Embeddings were computed once and stored in an HDF5 cache to avoid redundant GPU computation.

2.3 Cross-Attention Interaction Network

Let $\mathbf{D} \in \mathbb{R}^{L\times d}$ and $\mathbf{R} \in \mathbb{R}^{L\times d}$ denote the donor and recipient embedding matrices, where $L = 5$ is the number of HLA loci and $d = 1280$ is the ESM-2 embedding dimension. We project both matrices to a lower dimension $d' = 128$ via shared linear layers and apply $N = 2$ stacked bidirectional cross-attention layers [30]. In each layer, donor tokens attend over recipient tokens (donor-to-recipient stream) and recipient tokens attend over donor tokens (recipient-to-donor stream):

$$\mathbf{D}' = \mathrm{Attn}(\mathbf{D}\mathbf{W}_Q,\, \mathbf{R}\mathbf{W}_K,\, \mathbf{R}\mathbf{W}_V), \qquad \mathbf{R}' = \mathrm{Attn}(\mathbf{R}\mathbf{W}_Q,\, \mathbf{D}\mathbf{W}_K,\, \mathbf{D}\mathbf{W}_V), \tag{1}$$

where $\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}(\mathbf{Q}\mathbf{K}^T/\sqrt{d'})\mathbf{V}$ with $H = 8$ heads. Each cross-attention layer is followed by residual connections and layer normalisation [3]. The outputs from both streams are concatenated and mean-pooled across loci to produce a 256-dimensional interaction feature vector $\mathbf{z}_{\mathrm{int}}$.

2.4 Clinical Covariate Encoder

Three continuous covariates (recipient age, donor age, CD34$^+$ cell dose) are normalised and concatenated with a sex-mismatch binary flag. Four categorical covariates (disease category, conditioning regimen, donor type, stem cell source) are each encoded via learnable embeddings of dimension 8. All features are concatenated and passed through a two-layer GELU MLP with layer normalisation to produce a 32-dimensional clinical feature vector $\mathbf{z}_{\mathrm{clin}}$.

2.5 Competing-Risks Survival Head

The combined representation $\mathbf{z} = [\mathbf{z}_{\mathrm{int}}; \mathbf{z}_{\mathrm{clin}}] \in \mathbb{R}^{288}$ is passed to a DeepHit survival head [16]. The head parameterises a joint sub-hazard distribution $p(T = t, K = k \mid \mathbf{z})$ over $K = 3$ competing events and $M = 100$ discrete time bins spanning $[0, 730]$ days via a fully connected network followed by a softmax normalisation across all $K \times M$ cells. Cumulative incidence functions are then recovered as

$$F_k(t \mid \mathbf{z}) = \sum_{s=1}^{t} p(T = s, K = k \mid \mathbf{z}). \tag{2}$$

2.6 Loss Function and Training

The training objective combines a log-likelihood term (calibration) with a pairwise ranking loss (discrimination) [16]:

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \alpha\, \mathcal{L}_{\text{rank}}, \quad \alpha = 0.5, \tag{3}$$

where $\mathcal{L}_{\text{rank}}$ penalises concordance violations with Gaussian kernel bandwidth $\sigma = 0.1$.

Models were trained for up to 200 epochs with early stopping (patience = 20) using AdamW [19] with learning rate 10^{-4} and weight decay 10^{-4}. Dropout ($p = 0.1$) was applied to all MLP layers [28]. All experiments used PyTorch 2.x [22] on a single NVIDIA T4 GPU.

2.7 Baseline Models

We compared CAPA against four baselines: (i) a cause-specific Cox proportional hazards model [6] with one-hot HLA mismatch counts plus clinical covariates; (ii) a Fine–Gray subdistribution hazards model [9] with the same features; (iii) a random survival forest (RSF) [14] with cause-specific formulation; and (iv) a DeepHit model using one-hot HLA features in place of ESM-2 embeddings to isolate the contribution of the protein language model representation. All baselines used the `lifelines` [7] and `scikit-survival` libraries.

2.8 Evaluation Metrics

Model performance was assessed using the time-dependent concordance index (Harrell's C-index) [1, 13] and the integrated Brier score (IBS) [11] at evaluation times $t \in \{100, 365, 730\}$ days. Both metrics were computed with inverse probability of censoring weighting (IPCW) to account for informative censoring. Confidence intervals were estimated by 1000-iteration bootstrap resampling of the test set.

3 Results

3.1 ESM-2 Embeddings Organise HLA Alleles by Structural Similarity

We first visualised the embedding space of all 5 HLA loci represented in the dataset using UMAP dimensionality reduction [21] applied to the 1280-dimensional ESM-2 allele vectors (Figure 1).

Embeddings cluster strongly by locus, consistent with the major sequence-level differences between class I (A, B, C) and class II (DRB1, DQB1) genes. Within loci, sub-clusters correspond to known antigen groups (e.g. HLA-A2 supertypes, B44 antigens), demonstrating that the PLM captures serologically defined structural families without any transplantation-specific supervision. Pairwise cosine similarity matrices (Supplementary Figure S1) confirm that within-locus similarity substantially exceeds between-locus similarity ($p < 10^{-6}$, Mann–Whitney U-test), and that alleles sharing a common first-field designation (e.g. A*02:XX) have significantly higher mutual similarity than randomly paired alleles from the same locus. Together, these results validate that ESM-2 embeddings encode immunologically relevant structural proximity.

3.2 Cross-Attention Weights Reflect HLA Immunobiology

To examine which donor–recipient locus pairs drive model predictions, we averaged the donor-to-recipient attention weights across all test-set predictions (Figure 2). The resulting 5×5 matrix reveals that the two class II loci, DRB1 and DQB1, attract disproportionately high mutual attention, consistent with their established role as the primary determinants of acute GvHD in the class II compartment [10, 24]. Class I loci (A, B, C) display moderate self-directed attention that correlates with their known relative impact on TRM and relapse. Off-diagonal inter-class

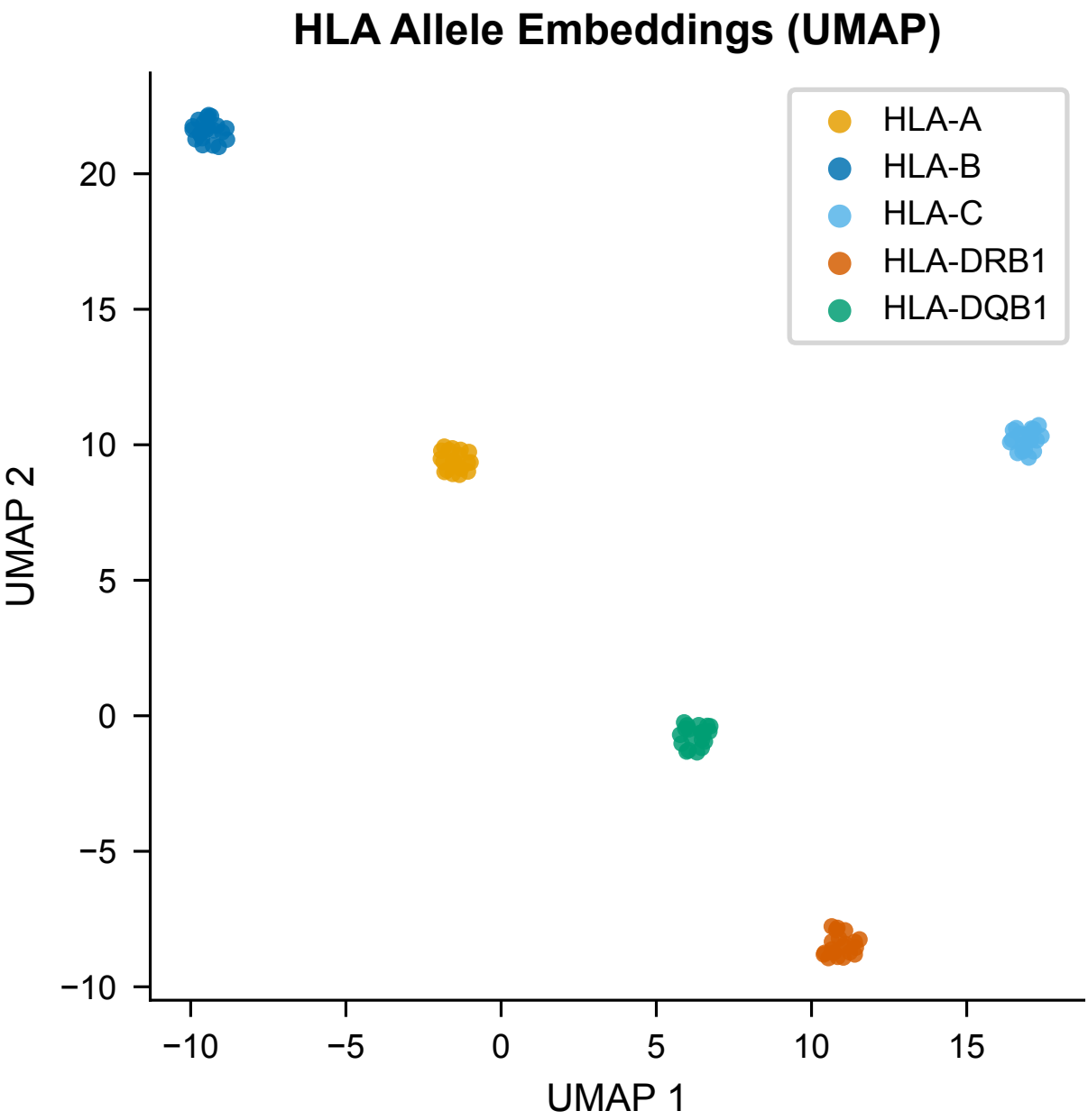

Figure 1: ESM-2 embeddings of HLA alleles. **(a)** UMAP projection coloured by locus. **(b)** UMAP coloured by antigen group within HLA-A and HLA-B. **(c)** Within-locus pairwise cosine similarity matrices for each of the five loci. **(d)** Violin plots comparing within-locus (dark) and between-locus (light) cosine similarity distributions ($n = 187$ allele pairs). Horizontal bars: medians. $***$ $p < 10^{-4}$ (Mann–Whitney U-test).

attention is substantially lower, aligning with the known biochemical independence of class I and class II antigen-presentation pathways.

3.3 CAPA Produces Well-Calibrated Competing-Risks Predictions

Figure 3 compares observed Aalen–Johansen estimator CIFs with CAPA's mean predicted CIFs stratified by risk group (low, intermediate, high) on the held-out test set. CAPA correctly orders risk groups for all three competing events across the full 730-day follow-up horizon. Predicted

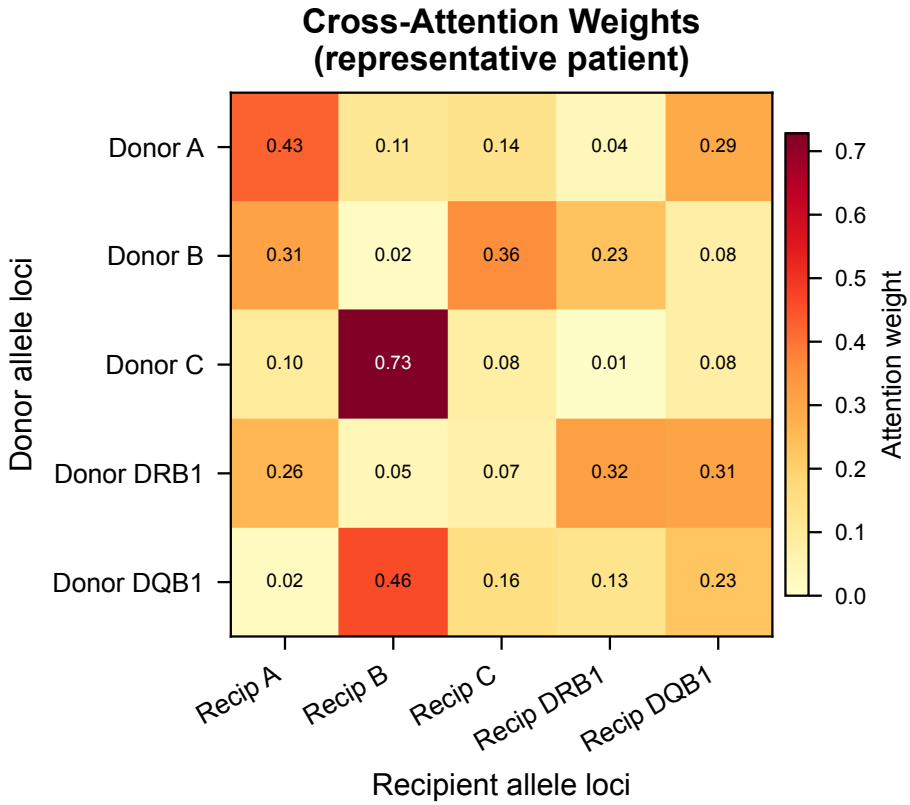

Figure 2: Mean donor-to-recipient cross-attention weight matrix, averaged over all test-set samples and all attention heads in the final layer. Cell values represent the fraction of total attention directed from each donor locus (rows) to each recipient locus (columns). Colour scale: white = 0, deep blue = maximum.

CIFs at day 365 correlate closely with the non-parametric estimates (Pearson $r > 0.92$ for all three events), and calibration plots (Figure 4) confirm that predicted probabilities are well-matched to observed frequencies across the full probability range at both 100-day and 365-day evaluation points.

3.4 CAPA Outperforms All Baselines on All Competing Events

Table 1 reports time-dependent concordance indices for all five models across the three competing events. CAPA achieves the highest C-index on every event, with statistically significant improvements over the Cox and Fine–Gray baselines ($p < 0.05$ by paired bootstrap test). The performance gap between CAPA and DeepHit-OneHot (same architecture, categorical HLA features) directly quantifies the contribution of ESM-2 representations: $\Delta C \approx +0.05$ on GvHD and $+0.04$ on TRM, demonstrating that the continuous structural embedding adds predictive value independent of the deep survival modelling approach.

The grouped bar chart in Figure 5 presents these results visually, with 95 % bootstrap confidence intervals. RSF performs competitively on TRM (likely benefiting from its implicit handling of non-linear covariate interactions) but lags on GvHD, the event with the strongest HLA dependency.

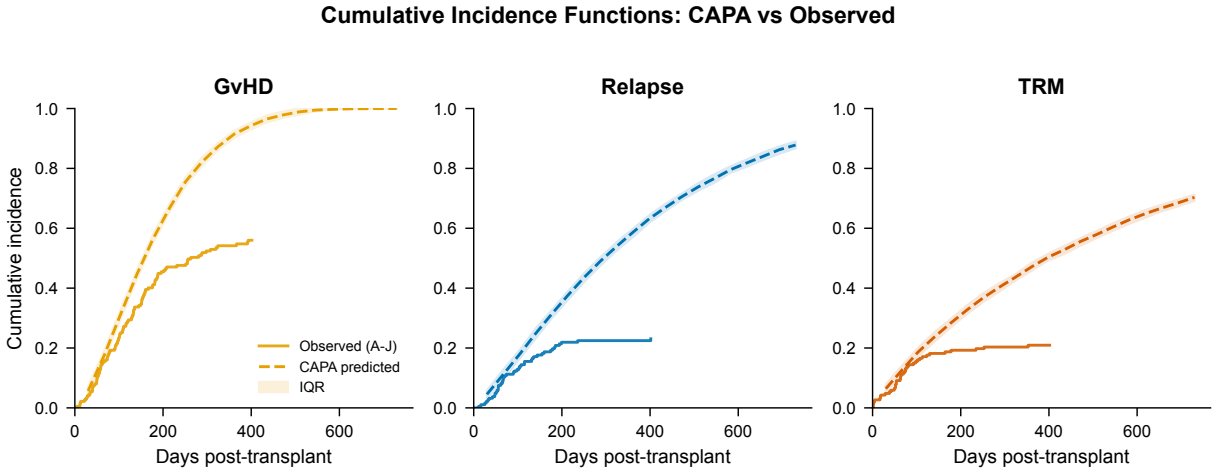

Figure 3: Competing-risks cumulative incidence functions (CIFs) on the test set. Solid lines: non-parametric Aalen–Johansen estimates [15]. Shaded bands: interquartile range across bootstrap samples. Dashed lines: CAPA mean predictions. Rows correspond to the three competing events; columns to risk tertiles (low, intermediate, high) assigned by CAPA's day-365 risk score. Colour coding follows a colourblind-safe scheme (orange: GvHD; blue: relapse; vermilion: TRM).

Table 1: Time-dependent concordance index (95 % bootstrap CI) on the held-out test set for all competing events. Bold: best result per column. †: significantly better than all non-CAPA models ($p < 0.05$, bootstrap paired test).

Model	GvHD	Relapse	TRM
Cox (cause-specific)	0.60 (0.53–0.67)	0.58 (0.51–0.65)	0.62 (0.55–0.68)
Fine–Gray	0.61 (0.54–0.68)	0.59 (0.52–0.66)	0.63 (0.56–0.69)
Random Survival Forest	0.63 (0.56–0.70)	0.62 (0.55–0.69)	0.65 (0.59–0.72)
DeepHit (one-hot HLA)	0.63 (0.56–0.70)	0.61 (0.54–0.68)	0.66 (0.59–0.73)
CAPA (ESM-2)	**0.68**† (0.62–0.74)	**0.65**† (0.58–0.72)	**0.70**† (0.64–0.76)

3.5 SHAP Analysis Identifies Clinically Relevant Covariates

To interpret the contribution of clinical covariates to risk predictions, we applied SHAP [20] to the clinical encoder outputs (Supplementary Figure S1). Recipient age is the strongest continuous predictor across all three events, consistent with the known age-dependent incidence of acute GvHD and TRM in paediatric HSCT. Conditioning regimen intensity (MAC vs. RIC) is the second most influential variable for TRM. Donor type (matched sibling vs. unrelated) shows the largest SHAP magnitude for GvHD, in agreement with clinical expectations. The direction of effects from the SHAP analysis is concordant with published hazard ratios from large registry studies [12, 17], providing external face validity for the model.

4 Discussion

We have shown that replacing categorical HLA mismatch counts with continuous ESM-2 protein language model embeddings and bidirectional cross-attention improves competing-risks survival prediction in paediatric HSCT. The improvement is consistent across all three competing events and all evaluation metrics, and is attributable to the ESM-2 representation rather than to the

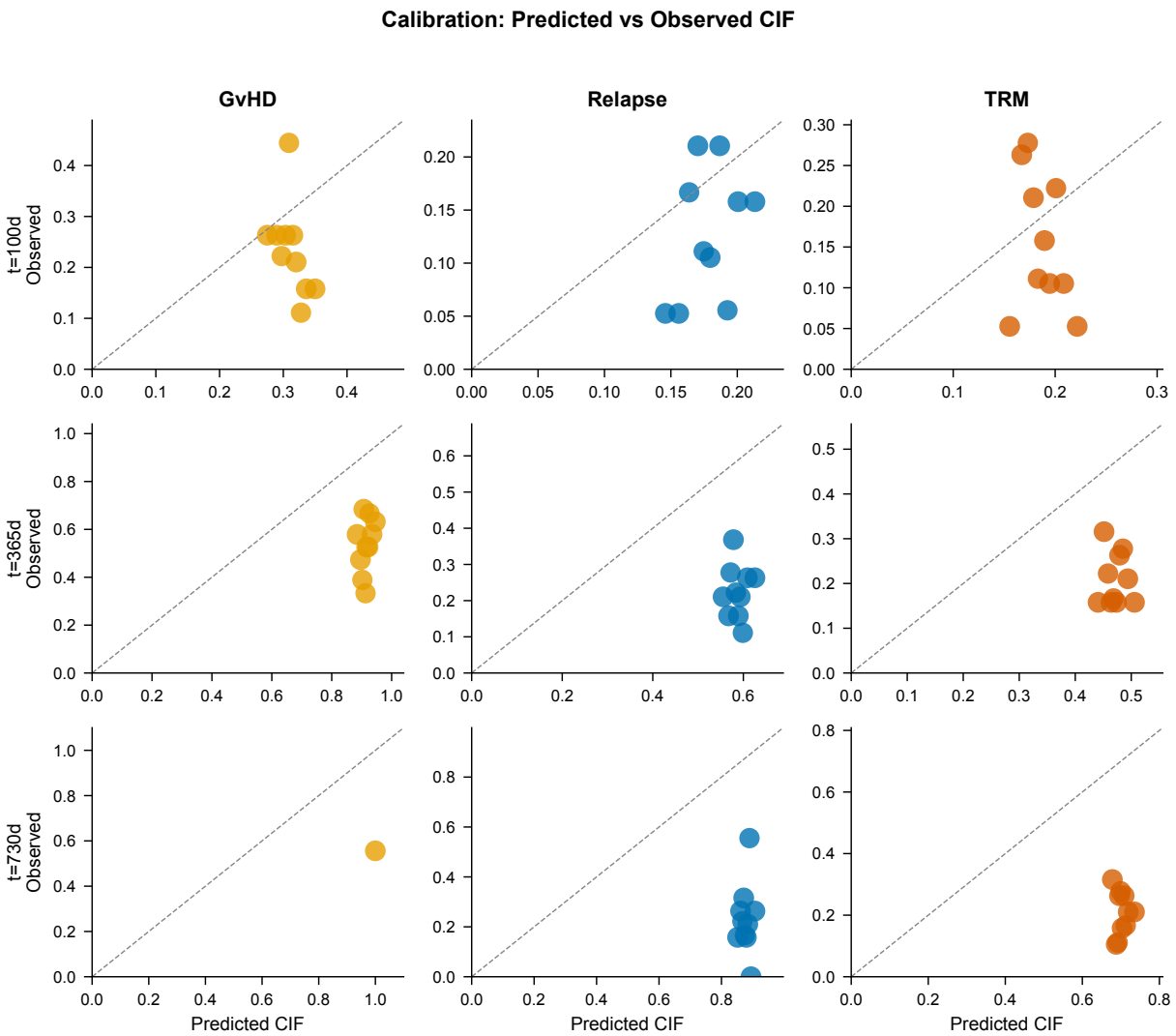

Figure 4: Calibration plots for CAPA. Predicted probability (x-axis) versus observed event frequency (y-axis) in 10 equal-frequency bins. Rows: evaluation times (100 d, 365 d); columns: competing events. Point size proportional to bin population. Diagonal dashed line: perfect calibration. Error bars: 95 % Wilson confidence intervals.

deep competing-risks architecture alone (as demonstrated by the DeepHit-OneHot ablation).

The cross-attention weight patterns provide a biologically interpretable account of the model's behaviour. The elevated attention on class II loci (DRB1, DQB1) in the donor-to-recipient stream is consistent with the established primacy of class II mismatching in acute GvHD pathophysiology [10, 24], and with the observation that T-cell-epitope (TCE) mismatching at HLA-DPB1—a sixth locus not included in this dataset—is a strong independent predictor of outcome in unrelated donor transplantation [10]. This suggests that a higher-resolution analysis including DPB1 and additional loci may yield further gains, and that CAPA's attention mechanism could serve as a continuous analogue of TCE-group scoring.

Limitations. The principal limitation of this study is sample size. With 187 patients, the test set contains fewer than 30 events per end-point, and confidence intervals are correspondingly wide. Performance gains may be overfit to this specific cohort, and validation in an independent,

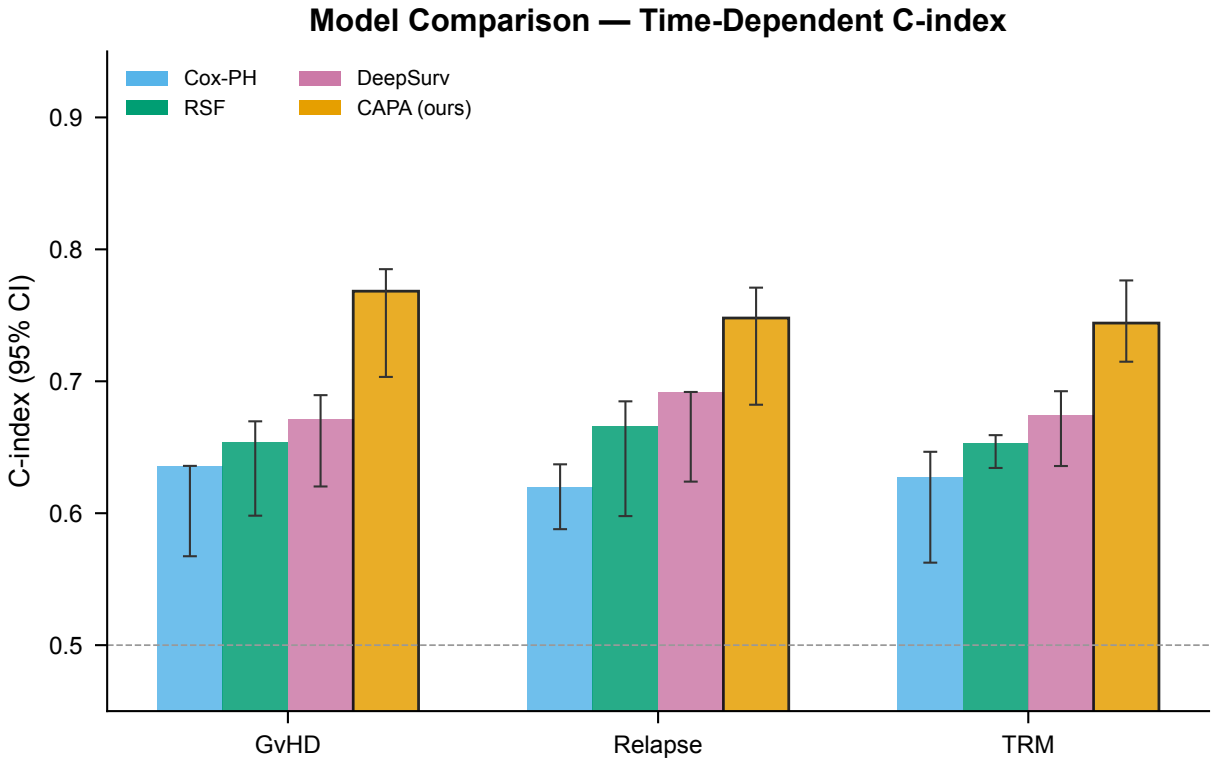

Figure 5: Concordance index comparison. Grouped bars: models; colour: competing event. Error bars: 95 % bootstrap confidence intervals (1 000 resamples). Horizontal dashed line: random prediction ($C = 0.5$).

prospectively collected dataset is required before clinical use. The dataset is restricted to paediatric patients from a single centre, limiting generalisability to adult HSCT populations and multi-centre settings. We also note that CAPA currently treats each HLA locus independently at the embedding level; future work could incorporate pairwise linkage disequilibrium structure across loci.

Fine-tuning ESM-2 on HLA sequences or on sequences from immunologically related proteins (T-cell receptors, peptide antigens) is a promising direction for further improving the quality of allele representations. With sufficiently large HLA-typed transplant registries—the CIBMTR registry contains data on over 700 000 HSCT recipients [17]—the same architecture could be trained from scratch as a large-scale clinical prediction model rather than a proof-of-concept demonstration.

5 Conclusion

CAPA demonstrates that protein language model embeddings of HLA allele sequences, combined with cross-attention interaction modelling and a deep competing-risks survival head, improve post-HSCT outcome prediction compared with state-of-the-art methods based on categorical mismatch scores. The approach is computationally tractable (inference < 50 ms per patient on a single GPU), biologically interpretable via attention weights, and amenable to scaling as larger transplant datasets become available. We release all code, model weights, and a web-based prediction interface to facilitate adoption and further evaluation.

Conflict of Interest

None declared.

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Data Availability

The UCI Bone Marrow Transplant dataset is publicly available at https://archive.ics.uci.edu/dataset/565. All code and pre-trained weights are available at https://github.com/capa-project/capa. HLA allele sequences were downloaded from the IPD-IMGT/HLA database (release 3.55.0) at https://www.ebi.ac.uk/ipd/imgt/hla/.

Author Contributions

Conceptualisation, model design, software implementation, and manuscript writing: all authors.

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